



Academic year (2026 - 2027)

SYNOPSIS

Project Title : ResilienceHub: AI-Enabled Disaster Response Platform

Title of Seminar : GIS and Google Maps in Emergency Response

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Class : 4N (IT)

Abstract:

Natural disasters such as floods, earthquakes, cyclones, fires, and pandemics require rapid and well-coordinated emergency response to minimize the loss of life and property. Geographic Information Systems (GIS) and Google Maps play a vital role in disaster management by providing real-time location-based information, improving situational awareness, and supporting effective decision-making during emergencies. This seminar explains the basic concepts of Geographic Information Systems (GIS) and Google Maps and their applications in emergency response. It discusses how GIS is used to identify disaster-affected areas, visualize hazard zones, map critical resources, and monitor emergency incidents through interactive dashboards. The seminar also explains the role of Google Maps API in live location tracking, shelter navigation, resource mapping, and optimized route planning for rescue teams and affected citizens. In addition, the seminar highlights the integration of GIS with disaster response platforms such as ResilienceHub, where emergency management agencies, hospitals, NGOs, volunteers, and disaster control authorities can coordinate rescue operations more effectively. It also discusses the use of weather services for early warning alerts, real-time environmental monitoring, and efficient resource allocation during disaster situations..

Keywords : Geographic Information System (GIS), Google Maps API, Emergency Response, Disaster Management, Live Location Tracking, Shelter Navigation, Resource Mapping, Real-Time Monitoring, Weather Alerts, Public Safety.

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